

# Jareen Anjom

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## RESEARCH BACKGROUND

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- **Deep Learning for Healthcare:** Leveraging deep learning models for medical and physiological data analysis, with applications in disease detection, health risk prediction, and continuous monitoring systems.
- **Edge Intelligence:** Developing lightweight, high-performance neural networks and exploring model optimization techniques to enable deployment on resource-constrained edge devices.
- **Computer Vision for Assistive Technologies:** Designing real-time perception systems, including object detection, semantic segmentation, and depth estimation, to enhance mobility and navigation.

## EDUCATION

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North South University, Bangladesh

Feb 2021 - Dec 2024

BSc Computer Science and Engineering | CGPA: 3.84/4

- Awarded 100% Merit Scholarship
- Graduated with *summa cum laude*
- Thesis: Visualization of Sorting Algorithms on a Mobile Application - A Study on Traditional vs Interactive Digital Tools, doi: [10.13140/RG.2.2.12458.17602](https://doi.org/10.13140/RG.2.2.12458.17602)
- Directed Research: Leveraging deep edge intelligence for real-time respiratory disease detection, doi: [10.1016/j.ceh.2025.01.001](https://doi.org/10.1016/j.ceh.2025.01.001)

## EXPERIENCE

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Research Assistant - Kuwait University | Kuwait

Jun 2025 - Present

*Project I: Heat Stress Prediction using Open-Source Wearables*

- Develop and optimize machine learning models for real-time heat stress prediction in the KU campus using physiological and environmental sensor data collected from Bangle.js 2 wearable devices.
- Review literature and draft research articles for journal and conference publications related to wearable health monitoring, heat stress prediction, and AI-based healthcare system

*Project II: Modeling Ultrasound Waves using Physics-Informed ResNet Approximators*

Research Grant: ZF01/25

June 2026 – Present

- Develop Physics-Informed Neural Networks (PINNs) and ResNet-based surrogate models for ultrasound wave propagation governed by partial differential equations (PDEs).
- Benchmark deep learning approximators against traditional Finite Difference Method (FDM) simulations.
- Extend modeling approaches toward realistic skull-image domains for biomedical ultrasound applications.

AI Specialist - Fidelity Technologies | Kuwait

May 2025 - Present

- Develop and maintain on-premise company AI assistant leveraging lightweight LLMs (HuggingFace, Ollama) for intelligent RAG (vector + bm25), and agents for multi task handling
- Integrate AI pipelines into existing content management systems to improve workload efficiency
- Engineer predictive machine learning algorithms improving company productivity with 90%+ accuracy

**Research Assistant** - North South University | Bangladesh

Jan 2024 - Jan 2025

Research Grant: CTRG-23-SEPS-29

- Performed original research in the field of computer vision & edge intelligence; designed and trained deep learning algorithms for specific objects of interest, solving real-world problems
- Tested, evaluated, and optimized models for resource-constrained devices

**Teaching Assistant** - North South University | Bangladesh

Jun 2023 - Dec 2024

- CSE311 Database Management System, assessed SQL/MS-SQL projects and assignments
- Conducted tutorial sessions for students needing extra help outside of class hours

## PUBLICATIONS

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### *Journal Articles*

- T. Hasan, **J. Anjom**, Md. I. A. Hossain, and Z. U. Shamszaman, "SatNet-B3: A Lightweight Deep Edge Intelligence Framework for Satellite Imagery Classification," *Future Internet*, vol. 17, no. 12, p. 579, Dec. 2025, doi: [10.3390/fi17120579](https://doi.org/10.3390/fi17120579)
- Md. I. A. Hossain, **J. Anjom**, and R. I. Chowdhury, "Towards Walkable Footpath Detection for the Visually Impaired on Bangladeshi Roads with Smartphones Using Deep Edge Intelligence," *Array*, Jul. 2025, doi: [10.1016/j.array.2025.100353](https://doi.org/10.1016/j.array.2025.100353)
- T. T. Oishee, **J. Anjom**, U. Mohammed, and Md. I. A. Hossain, "Leveraging deep edge intelligence for real-time respiratory disease detection," *Clinical eHealth*, Jan. 2025, doi: [10.1016/j.ceh.2025.01.001](https://doi.org/10.1016/j.ceh.2025.01.001)
- R. I. Chowdhury, **J. Anjom**, and Md. I. A. Hossain, "A Novel Edge Intelligence-based Solution for Safer Footpath Navigation of Visually Impaired Using Computer Vision," *Journal of King Saud University - Computer and Information Sciences*, vol. 36, no. 8, p. 102191, Sep. 2024, doi: [10.1016/j.jksuci.2024.102191](https://doi.org/10.1016/j.jksuci.2024.102191)

### *Conference Proceedings*

- **J. Anjom**, R. I. Chowdhury, T. Hasan, and Md. I. A. Hossain, "An Embedded Real-time Object Alert System for Visually Impaired: A Monocular Depth Estimation based Approach through Computer Vision," Accepted - 2025 IEEE International Conference on Advanced Computing Technologies (ICoACT-2025), doi: [10.48550/arXiv.2507.08165](https://doi.org/10.48550/arXiv.2507.08165)
- M. N. Mantaqa, **J. Anjom**, and Md. I. A. Hossain, "Diabetic Retinopathy Detection Using a Lightweight Edge Intelligence-based Technique," in 2024 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON 2024), Nov. 2024, pp. 113–118, doi: [10.1109/BECITHCON64160.2024.10962647](https://doi.org/10.1109/BECITHCON64160.2024.10962647)

## ONGOING RESEARCH

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- **J. Anjom**, A. Elshabrawi, S. Alkhader, and Z. Marafie, "Heat Stress Prediction using ML Methods on Environmental and Physiological Features in a Hot-Arid Climate"
- **J. Anjom**, T. T. Oishee, U. Mohammed, and Md. I. A. Hossain, "Drishti - Bangla Braille Character Recognition System using Deep Edge Intelligence"

## SKILLS

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**Programming Languages:** Python, Java, Kotlin

**AI & Machine Learning:** TensorFlow, PyTorch, Pandas

**LLMs:** RAG, Ollama, HuggingFace

**Databases:** MySQL, MS-SQL, SQLite, PostGres

**Others:** Git, MS Office, LaTeX

**Languages:** English (IELTS 7.5), Bangla, Hindi